

QUESTIONS 31 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner s1=new Scanner(System.in);
5         int a;
6         int b;
7         System.out.println("enter a number:");
8         a=s1.nextInt();
9         b=s1.nextInt();
10        System.out.println("addition="+ (a+b) + "subtraction=" + (a-b) + "multiplication=" + (a*b));
11    }
12 }
13 }
```

OUTPUT

```
enter a number:
addition=30subtraction=-10multiplication=200
```

INPUT

10
20

QUESTIONS 31 / 31 completed

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner s1=new Scanner(System.in);
5         int a;
6         int b;
7         int c;
8         System.out.println("enter three number:");
9         a=s1.nextInt();
10        b=s1.nextInt();
11        c=s1.nextInt();
12        if(a>b && a>c){
13            System.out.println("maximum="+a);
14        }
15        else if(b>a && b>c){
16            System.out.println("maximum="+b);
17        }
18        else
19            System.out.println("maximum="+c);
20    }
21 }
```

OUTPUT

```
enter three number:
maximum=15
```

INPUT

15
5
9

QUESTIONS 31 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner s1=new Scanner(System.in);
5         int y;
6         System.out.println("enter a year:");
7         y=s1.nextInt();
8         if((y%4==0 && y%100!=0)|| (y%400==0)){
9             System.out.println(y+"is a leap year");
10        }else{
11            System.out.println(y+"is not a leap year");
12        }
13    }
14 }
```

OUTPUT

```
enter a year:
2025is not a leap year
```

INPUT

2025

QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int a = 0, b = 1;
7         while (a <= n) {
8             System.out.print(a + " ");
9             int c = a + b;
10            a = b;
11            b = c;
12        }
13    }
14 }
15 }
```

OUTPUT

```
0 1 1 2 3 5
```

INPUT

5

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

INPUT

5

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int fact = 1;
7         for (int i = 1; i <= n; i++) {
8             fact = fact * i;
9         }
10        System.out.println(fact);
11    }
12 }
```

OUTPUT

120

QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:

```
**
***
****
*****
```

PROGRAM EDITOR

Java

Run

Save

INPUT

Your INPUT
here!

```
1 public class StarPattern {
2     public static void main(String[] args) {
3         for (int i = 1; i <= 5; i++) {
4             for (int j = 1; j <= i; j++) {
5                 System.out.print("*");
6             }
7             System.out.println();
8         }
9     }
10 }
11 }
```

OUTPUT

```
*
**
***
****
```

QUESTIONS 31 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner; class Main {
2     public static void main(String[] args) {
3         Scanner sc = new Scanner(System.in);
4         int n = sc.nextInt();
5         int[] arr = new int[n];
6         for (int i = 0; i < n; i++) {
7             arr[i] = sc.nextInt();
8         }
9         int key = sc.nextInt();
10
11         int low = 0;
12         int high = n - 1;
13         int found = 0;
14         while (low <= high) {
15             int mid = (low + high) / 2;
16             if (arr[mid] == key) {
17                 System.out.println("Element found at index " + mid);
18                 found = 1;
19                 break;
20             } else if (arr[mid] < key) {
21                 low = mid + 1;
```

OUTPUT

Element found at index 2

INPUT

5
10 20 30 40 50
30

QUESTIONS 31 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int a = sc.nextInt();
6         int b = sc.nextInt();
7         int gcd = 1;
8         for (int i = 1; i <= a && i <= b; i++) {
9             if (a % i == 0 && b % i == 0) {
10                 gcd = i;
11             }
12         }
13         System.out.println(gcd);
14     }
15 }
```

OUTPUT

6

INPUT

12
18

QUESTIONS 31 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int count = 0;
7         for (int i = 1; i <= n; i++) {
8             if (n % i == 0) {
9                 count++;
10            }
11        }
12        if (count == 2)
13            System.out.println("Prime");
14        else
15            System.out.println("Not Prime");
16    }
17 }
```

OUTPUT

Not Prime

INPUT

4

QUESTIONS 31 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] a = new int[n];
7         for (int i = 0; i < n; i++) {
8             a[i] = sc.nextInt();
9         }
10        for (int i = 0; i < n - 1; i++) {
11            int min = i;
12            for (int j = i + 1; j < n; j++) {
13                if (a[j] < a[min]) {
14                    min = j;
15                }
16            }
17            int temp = a[i];
18            a[i] = a[min];
19            a[min] = temp;
20        }
21        for (int i = 0; i < n; i++) {
```

OUTPUT

11 12 22 25 64

INPUT

5
64 25 12 22 11

QUESTIONS 31 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         for (int i = 1; i <= 10; i++) {
7             System.out.println(n + " x " + i + " = " + (n * i));
8         }
9     }
10 }
```

OUTPUT

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
```

INPUT

5

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

Java

✓ Updated

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     static int factorial(int n) {
4         if (n == 0 || n == 1)
5             return 1;
6         else
7             return n * factorial(n - 1);
8     }
9     public static void main(String[] args) {
10         Scanner sc = new Scanner(System.in);
11         int n = sc.nextInt();
12         System.out.println(factorial(n));
13     }
14 }
```

OUTPUT

```
120
```

INPUT

5

QUESTIONS 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

**
*

PROGRAM EDITOR

Java

Run

Save

```
1 class Main {
2     public static void main(String[] args) {
3
4         for (int i = 5; i >= 1; i--) {
5
6             for (int j = 1; j <= i; j++) {
7                 System.out.print("*");
8             }
9
10            System.out.println();
11        }
12    }
13 }
```

OUTPUT

**
*

INPUT

Your INP
here!

QUESTIONS 31 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int a = sc.nextInt();
6         int b = sc.nextInt();
7         int lcm;
8         if (a > b)
9             lcm = a;
10        else
11            lcm = b;
12        while (true) {
13            if (lcm % a == 0 && lcm % b == 0) {
14                System.out.println(lcm);
15                break;
16            }
17            lcm++;
18        }
19    }
20 }
```

OUTPUT

36

INPUT

12
18

QUESTIONS 31 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String args[]){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int binary=0;
7         int place =1;
8         while(n>0){
9             int rem=n%2;
10            binary +=(rem*place);
11            place=place*10;
12            n=n/2;
13        }
14        System.out.println(binary);
15    }
16 }
```

OUTPUT

1010

INPUT

10

QUESTIONS 31 / 31 completed

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String args[]){
4         Scanner sc=new Scanner(System.in);
5         int binary=sc.nextInt();
6         int decimal=0;
7         int place=1;
8         while (binary>0){
9             int rem=binary%10;
10            decimal +=(rem*place);
11            place=place*2;
12            binary=binary/10;
13        }
14        System.out.println(decimal);
15    }
16 }
```

OUTPUT

10

INPUT

1010

QUESTIONS 31 / 31 completed

Sum ▾

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java ▾

Run

Save

INPUT

1234

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int sum = 0;
7         while (n > 0) {
8             sum = sum + (n % 10);
9             n = n / 10;
10        }
11        System.out.println("Sum of Digits:"+sum);
12    }
13 }
```

OUTPUT

Sum of Digits:10

QUESTIONS 31 / 31 completed

Product ▾

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java ▾

✓ Updated

Run

Save

INPUT

1234

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int product = 1;
7         while (n > 0) {
8             product = product * (n % 10);
9             n = n / 10;
10        }
11        System.out.println(product);
12    }
13 }
```

OUTPUT

24

QUESTIONS 31 / 31 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java

✓ Updated

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int base = sc.nextInt();
6         int exponent = sc.nextInt();
7         int power = 1;
8         for (int i = 1; i <= exponent; i++) {
9             power = power * base;
10        }
11        System.out.println(power);
12    }
13 }
```

INPUT

2
5

OUTPUT

32

QUESTIONS 31 / 31 completed

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         double radius = sc.nextDouble();
6         double area = 3.14 * radius * radius;
7         System.out.println(area);
8     }
9 }
```

INPUT

5

OUTPUT

78.5

QUESTIONS 31 / 31 completed

Sphere

SPHERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

Java

✓ Updated

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         double radius = sc.nextDouble();
6         double volume = (4.0 / 3.0) * 3.14 * radius * radius * radius;
7         System.out.println(volume);
8     }
9 }
```

OUTPUT

113.03999999999998

INPUT

3

QUESTIONS 31 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java

✓ Updated

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         String rev = "";
7         for (int i = str.length() - 1; i >= 0; i--) {
8             rev = rev + str.charAt(i);
9         }
10        if (str.equals(rev))
11            System.out.println("Palindrome");
12        else
13            System.out.println("Not Palindrome");
14    }
15 }
```

OUTPUT

Palindrome

INPUT

madam

QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java

✓ Updated

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int temp = n;
7         int sum = 0;
8         while (n > 0) {
9             int digit = n % 10;
10            sum = sum + (digit * digit * digit);
11            n = n / 10;
12        }
13        if (sum == temp)
14            System.out.println("Armstrong Number");
15        else
16            System.out.println("Not Armstrong Number");
17    }
18 }
19 }
```

OUTPUT

Armstrong Number

INPUT

153

QUESTIONS 31 / 31 completed

RT-num.Pattern

RT- NUM.PATTERN

Write a Java program to print the following pattern using loops:

```
1
22
333
4444
55555
```

PROGRAM EDITOR

Java

Run

Save

```
1 class Main {
2     public static void main(String[] args) {
3         for (int i = 1; i <= 5; i++) {
4             for (int j = 1; j <= i; j++) {
5                 System.out.print(i);
6             }
7             System.out.println();
8         }
9     }
10 }
```

OUTPUT

```
1
22
333
4444
55555
```

INPUT

Your INPUT here!

QUESTIONS 31 / 31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int sum = 0;
7         for (int i = 1; i < n; i++) {
8             if (n % i == 0) {
9                 sum = sum + i;
10            }
11        }
12        if (sum == n)
13            System.out.println("Perfect Number");
14        else
15            System.out.println("Not Perfect Number");
16    }
17 }
```

OUTPUT

Not Perfect Number

INPUT

5

QUESTIONS 31 / 31 completed

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] a = new int[n];
7         for (int i = 0; i < n; i++) {
8             a[i] = sc.nextInt();
9         }
10        int max = a[0];
11        for (int i = 1; i < n; i++) {
12            if (a[i] > max) {
13                max = a[i];
14            }
15        }
16        System.out.println(max);
17    }
18 }
```

OUTPUT

40

INPUT

5
10 25 8 40 15

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        int largest = Integer.MIN_VALUE;
11        int secondLargest = Integer.MIN_VALUE;
12        for (int i = 0; i < n; i++) {
13            if (arr[i] > largest) {
14                secondLargest = largest;
15                largest = arr[i];
16            } else if (arr[i] > secondLargest && arr[i] != largest) {
17                secondLargest = arr[i];
18            }
19        }
20        System.out.println(secondLargest);
21    }
```

OUTPUT

30

INPUT

5
10 25 8 40 30

QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        int min = arr[0];
11
12        for (int i = 1; i < n; i++) {
13            if (arr[i] < min) {
14                min = arr[i];
15            }
16        }
17        System.out.println(min);
18    }
19 }
```

OUTPUT

8

INPUT

5
10 25 8 40 30

QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        int smallest = Integer.MAX_VALUE;
11        int secondSmallest = Integer.MAX_VALUE;
12        for (int i = 0; i < n; i++) {
13            if (arr[i] < smallest) {
14                secondSmallest = smallest;
15                smallest = arr[i];
16            } else if (arr[i] < secondSmallest && arr[i] != smallest) {
17                secondSmallest = arr[i];
18            }
19        }
20        System.out.println(secondSmallest);
21    }
```

OUTPUT

10

INPUT

5
10 25 8 40 30

QUESTIONS 31 / 31 completed

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        for (int i = 0; i < n - 1; i++) {
11            for (int j = 0; j < n - 1 - i; j++) {
12                if (arr[j] > arr[j + 1]) {
13                    int temp = arr[j];
14                    arr[j] = arr[j + 1];
15                    arr[j + 1] = temp;
16                }
17            }
18        }
19        for (int i = 0; i < n; i++) {
20            System.out.print(arr[i] + " ");
21        }
```

OUTPUT

12 22 25 34 64

INPUT

5
64 34 25 12 22

QUESTIONS 31 / 31 completed

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        for (int i = 1; i < n; i++) {
11            int key = arr[i];
12            int j = i - 1;
13            while (j >= 0 && arr[j] > key) {
14                arr[j + 1] = arr[j];
15                j--;
16            }
17            arr[j + 1] = key;
18        }
19        for (int i = 0; i < n; i++) {
20            System.out.print(arr[i] + " ");
21        }
22    }
23 }
```

OUTPUT

12 22 25 34 64

INPUT

5
64 34 25 12 22